

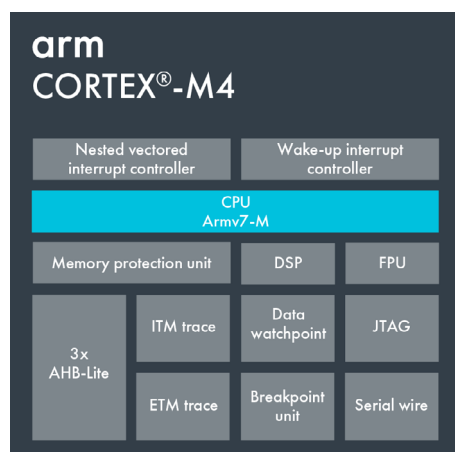


Figure 1: Block diagram of the Cortex-M4 processor

Overview

The Cortex-M4 processor is developed to address digital signal control markets that demand an efficient, easy-to-use blend of control and signal processing capabilities. The combination of high-efficiency signal processing functionality with the low-power, low-cost and ease-of-use benefits of the Cortex-M family of processors satisfies many markets. These industries include motor control, automotive, power management, embedded audio and industrial automation markets.

Features

Feature	Description
Architecture	Armv7E-M
Bus Interface	3x AMBA AHB-Lite interface (Harvard bus architecture) AMBA ATB interface for CoreSight debug components
ISA Support	Thumb/Thumb-2
Pipeline	3-stage + branch speculation
DSP Extension	Single-cycle 16/32-bit MAC Single-cycle dual 16-bit MAC 8/16-bit SIMD arithmetic Hardware Divide (2-12 cycles)
Floating-Point Unit	Optional single precision floating point unit (FPU) IEEE 754 compliant
Memory Protection	Optional 8-region MPU with sub regions and background region
Bit Manipulation	Integrated Bit Field Processing Instructions & Bus Level Bit Banding
Interrupts	Non-maskable interrupt (NMI) + 1 to 240 physical interrupts
Interrupt Priority Levels	8 to 256 priority levels
Wake-up Interrupt Controller	Optional
Sleep Modes	Integrated WFI and WFE Instructions and Sleep On Exit capability Sleep & Deep Sleep Signals Optional Retention Mode with Arm Power Management Kit
Debug	Optional JTAG and Serial Wire Debug ports Up to 8 Breakpoints and 4 Watchpoints
Trace	Optional Instruction Trace (ETM), Data Trace (DWT), and Instrumentation Trace (ITM)

About the Processor

The Cortex-M4 processor is a low-power processor that features low gate count, low interrupt latency, and low-cost debug. The Cortex-M4 with FPU is a processor with the same capability as the Cortex-M4 processor and includes floating-point arithmetic functionality. Both processors are intended for deeply embedded applications that require fast interrupt response features.

The Cortex-M4 processor includes:

- ✦ A processor core
- ✦ A Nested Vectored Interrupt Controller (NVIC) closely integrated with the processor core to achieve low latency interrupt processing
- ✦ Multiple high-performance bus interfaces
- ✦ A low-cost debug solution with the optional ability to:
 - Implement breakpoints and code patches
 - Implement watchpoints, tracing, and system profiling
 - Support printf style debugging
 - Bridge to a Trace Port Analyzer (TPA)
- ✦ An optional Memory Protection Unit (MPU)
- ✦ A Floating Point Unit (FPU) in the Cortex-M4 with FPU processor

Block Diagram

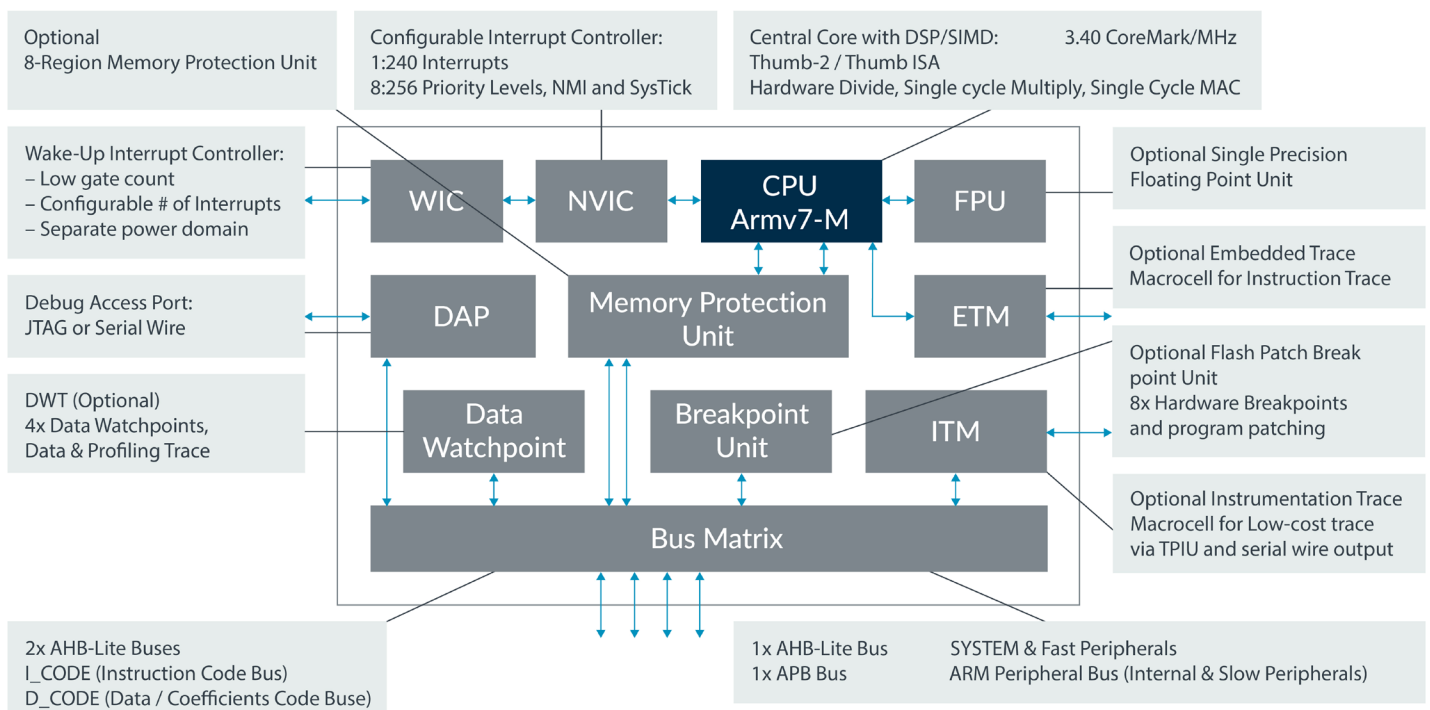


Figure 2: Cortex-M4 processor components

Cortex-M4 Components

Processor

The Cortex-M4 processor features a low gate count processor core, with low latency interrupt processing that has:

- + A subset of the Thumb instruction set, defined in the Armv7-M architecture
- + Banked Stack Pointer (SP)
- + Hardware divide instructions, SDIV and UDIV
- + Handler and Thread modes
- + Thumb and Debug states
- + Support for interruptible-continued instructions LDM, STM, PUSH, and POP for low interrupt latency
- + Automatic processor state saving and restoration for low latency Interrupt Service Routine (ISR) entry and exit
- + Support for Armv6 big-endian byte-invariant or little-endian accesses
- + Support for Armv6 unaligned accesses

Floating Point Unit (FPU) in the Cortex-M4 with FPU processor providing:

- + 32-bit instructions for single-precision (C float) data-processing operations
- + Combined Multiply and Accumulate instructions for increased precision (Fused MAC)
- + Hardware support for conversion, addition, subtraction, multiplication with optional accumulate, division, and square root
- + Hardware support for denormals and all IEEE rounding modes
- + 32 dedicated 32-bit single precision registers, also addressable as 16 double-word registers
- + Decoupled three-stage pipeline

Nested Vectored Interrupt Controller

Nested Vectored Interrupt Controller (NVIC) closely integrated with the processor core to achieve low latency interrupt processing. Features include:

- + External interrupts, configurable from 1 to 240
- + Bits of priority, configurable from 3 to 8
- + Dynamic reprioritization of interrupts
- + Priority grouping
 - This enables selection of preempting interrupt levels and non preempting interrupt levels.
- + Support for tail-chaining and late arrival of interrupts
 - This enables back-to-back interrupt processing without the overhead of state saving and restoration between interrupts.

- + Processor state automatically saved on interrupt entry, and restored on interrupt exit, with no instruction overhead
- + Optional Wake-up Interrupt Controller (WIC), providing ultra-low power sleep mode support

Memory Protection Unit (MPU)

An optional MPU for memory protection, including:

- + Eight memory regions
- + Sub Region Disable (SRD), enabling efficient use of memory regions
- + The ability to enable a background region that implements the default memory map attributes

Cross Trigger Interface Unit

The optional CTI enables the debug logic and the Embedded Trace Macrocell (ETM) to interact with each other and with other CoreSight components.

Debug and Trace

Low-cost debug solution that features:

- + Debug access to all memory and registers in the system, including access to memory mapped devices, access to internal core registers when the core is halted, and access to debug control registers even while SYSRESETn is asserted
- + Serial Wire Debug Port (SW-DP) or Serial Wire JTAG Debug Port (SWJ-DP) debug access, or both
- + Optional Flash Patch and Breakpoint (FPB) unit for implementing breakpoints and code patches
- + Optional Data Watchpoint and Trace (DWT) unit for implementing watchpoints, data tracing, and system profiling
- + Optional Instrumentation Trace Macrocell (ITM) for support of printf style debugging
- + Optional Trace Port Interface Unit (TPIU) for bridging to a Trace Port Analyzer (TPA), including Single Wire Output (SWO) mode
- + Optional ETM for instruction trace

ETM interface

The ETM interface enables simple connection of an ETM to the processor. It provides a channel for instruction trace to the ETM.

AHB Trace Macrocell interface

The AHB Trace Macrocell (HTM) interface enables a simple connection of the AHB trace macrocell to the processor. It provides a channel for the data trace to the HTM.

Your implementation must include this interface to use the HTM interface. You must set TRCENA to 1 in the Debug Exception and Monitor Control Register (DEMCR) before you enable the HTM to enable the HTM port to supply trace data.

Bus interfaces

- + Three **Advanced High-performance Bus-Lite** (AHB-Lite) interfaces: ICode, DCode, and System bus interfaces
- + Private Peripheral Bus (PPB) based on **Advanced Peripheral Bus (APB) interface**
- + **Bit-band support** that includes atomic bit-band write and read operations
- + Memory access alignment
- + **Write buffer for buffering of write data**
- + Exclusive access transfers for multiprocessor systems

Debug port AHB-AP interface

The processor contains an **Advanced High-performance Bus Access Port (AHB-AP) interface** for debug accesses. An external Debug Port (DP) component accesses this interface. The Cortex-M4 system supports three possible DP implementations:

- + Serial Wire JTAG Debug Port (SWJ-DP)
 - The SWJ-DP is a standard CoreSight debug port that combines JTAG-DP and Serial Wire Debug Port (SW-DP).
- + SW-DP
 - This provides a two-pin interface to the AHB-AP port.
- + No DP present
 - If no debug functionality is present within the processor, a DP is not required.

The two DP implementations provide different mechanisms for debug access to the processor. Your implementation must contain only one of these components.

Cortex-M4 Pipeline

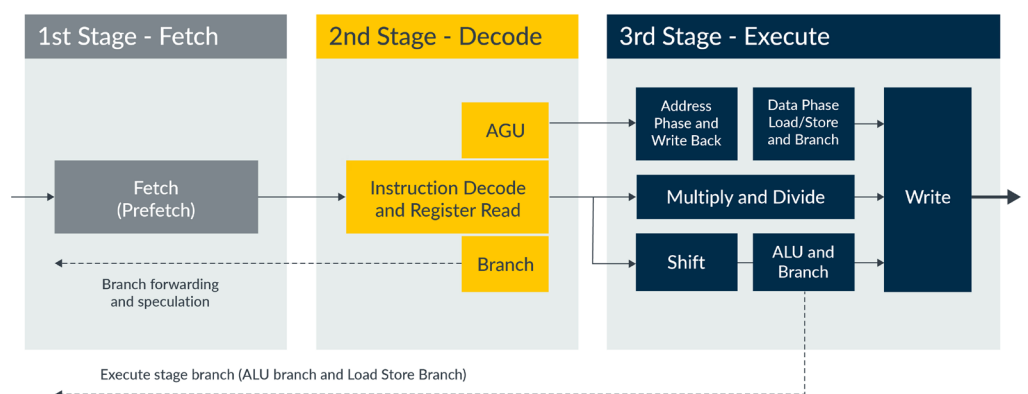


Figure 3: Cortex-M4 processor pipeline

Processor Configuration Options

The Cortex-M4 processor has configurable options that you can set during the implementation and integration stages to match your functional requirements.

Feature	Options
Number of interrupts	Specifies number of interrupts (1-240 interrupts)
Levels of interrupt priority	Between 3 and 8 bits of interrupt priority, between 8 and 256 levels of priority
MPU present	No MPU
	MPU present
FPU present	No FPU
	FPU present
Bit Banding	Specifies whether bit-banding is present
AHB Control	Specifies whether AB-Lite buses maintain control information during wait stated transfers
Debug Level	Specifies the level of debug support (0 to 3)
Trace Level	Specifies the level of trace support (0 to 3)
Reset all registers	Specifies whether all synchronous state or only architecturally required state is reset
JTAG present	Enables or disables the JTAG portion of the debug port
Clock Gate Present	Specifies whether architectural clock gates are included
Observation	Enables observation of internal state of the processor
WIC present	Specifies whether a WIC is present
WIC Lines	Number of WIC lines (minimum 2)

Instruction Set

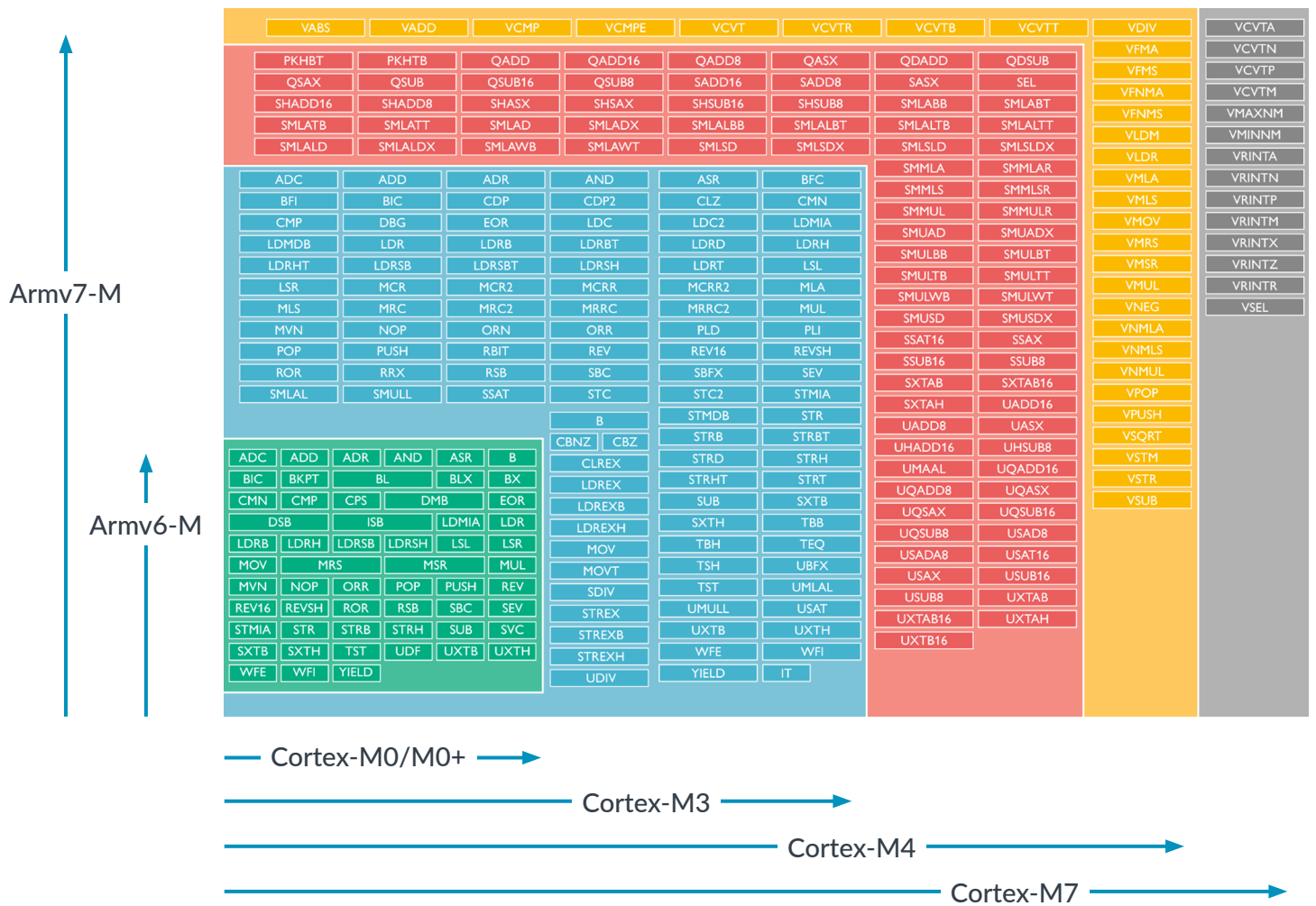


Figure 5: Instruction set

Power, Performance and Area

DMIPS	CoreMark/MHz
1.25	3.42

Configuration	90LP (7-track, RVt, typical 1.2V, 25°C)		65LP (8 Track , RVt, typical 1.2V, 25°C)		40LP (9 Track, RVt, typical 1.1v, 25°C)		28HPM (9-track, HVt, typical 0.9v, 85°C)	
	Area mm ²	Power μW/MHz	Area mm ²	Power μW/MHz	Area mm ²	Power μW/MHz	Area mm ²	Power μW/MHz
Minimum Configuration*	0.119	32.82	0.076	26.40	0.028	12.26	0.020	8.47
Feature Rich**	0.304	41.47	0.201	35.54	0.082	15.48	0.053	11.20

Max Freq	40LP (9-track RVt, typical 1.1v, 25C)	28HPM (12track , LVt, typical 0.9v, 85°C)
Feature Rich Configuration**	223MHz	822MHz

* MPU_PRESENT 0; NUM_IRQ 1; LVL_WIDTH 3; TRACE_LVL 0; DEBUG_LVL 0; JTAG_PRESENT 0; CLKGATE_PRESENT 1; RESET_ALL_REGS 0; WIC_PRESENT 0; WIC_LINES 3; BB_PRESENT 0; CONST_AHB_CTRL 0; FPU_PRESENT 0;

** MPU_PRESENT 1; NUM_IRQ 64; LVL_WIDTH 3; TRACE_LVL 2; DEBUG_LVL 3; JTAG_PRESENT 0; CLKGATE_PRESENT 1; RESET_ALL_REGS 0; WIC_PRESENT 1; WIC_LINES 67; BB_PRESENT 0; CONST_AHB_CTRL 0; FPU_PRESENT 1;

Additional Technical documents

1. Cortex-M4 Technical Reference Manual - [TRM](#)
2. Cortex-M4 Integration and Implementation Manual – available as part of the Bill of Materials
3. Armv7-M Architecture Reference Manual - [ARM](#)
4. CoreSight ETM-M4 Technical Reference Manual - [ETM](#)

Glossary of Terms

AHB	Advanced High-performance Bus
ATB	Advanced Trace Bus
C-AHB	Code AHB
CTI	Cross Trigger Interface
D-AHB	Debug AHB
DSP	Digital Signal Processing
DWT	Data Watchpoint and Trace
ETM	Embedded Trace Macrocell
FPU	Floating Point Unit
IEEE	Institute of Electrical and Electronics Engineers
ISR	Interrupt Service Routine
ITM	Instrumentation Trace Macrocell
JTAG	Joint Test Action Group
MAC	Multiply and Accumulate
MPU	Memory Protection Unit
NMI	Non-maskable Interrupt
NVIC	Nested Vectored Interrupt Controller
PPB	Private Peripheral Bus
S-AHB	System AHB
SIMD	Single Instruction, Multiple Data
SWO	Serial Wire Output
TPA	Trace Port Analyzer
TPIU	Trace Port Interface Unit
WFE	Wait for event
WFI	Wait for interrupt
WIC	Wake-up Interrupt Controller

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